Where the Signal Is

Label provenance and the limits of fusion in Arabic multimodal affect recognition

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What we set out to test (from RP1)

In RP1 we proposed **Tri-CAN**: text features query the audio and video features through cross-attention before classification.

The argument was that late fusion (encode each modality separately, concatenate at the end) cannot catch cases where the words and the delivery disagree, such as sarcasm.

Question for RP2: does cross-modal attention beat late fusion, and if it does, is the advantage concentrated on samples where the modalities disagree?

Baseline to beat

Gated late fusion on EGY-MER
weighted F1 = **0.519**
Hussein et al., 2025

Encoders

Text	AraBERT
Audio	wav2vec2
Video	Swin / ViT

Two things changed after the proposal

EGY-MER arrived late

We requested it and heard nothing back for months, so we built the pipeline on **Ar-MuSA**, the one openly downloadable Arabic tri-modal corpus (8,699 clips, 3-class sentiment).

EGY-MER arrived on 9 August. Everything was then repeated on it.

Part of the RP1 gap was closed by others

Our systematic search (1,322 papers) found that cross-modal attention reached Arabic *sentiment* in 2026 (GCMA-Net and two related papers, all on Ar-MuSA).

It has still not been applied to Arabic *emotion* recognition, which is what EGY-MER is.

Both changes pushed the project the same way: instead of just reporting one more F1 score, we had to understand what these benchmarks can and cannot tell us.

Ben Cheikhi et al., 2026; Khaled et al., 2025

How the experiments were run

- Encoders are **frozen**; features are extracted once and cached. Only the fusion module and classifier are trained.
- So when two models are compared, the fusion mechanism is the only thing that differs.
- Stratified 5-fold cross-validation, **identical folds** for every model.
- Predictions are collected out-of-fold, so every sample is predicted by a model that never trained on it.
- EGY-MER results are the mean of **3 seeds**. A difference smaller than the seed standard deviation is treated as a tie.
- Main metric: macro-F1 (the classes are very imbalanced). Weighted F1 only for comparing with the published baseline.
- Controls: a late-fusion model widened to Tri-CAN's parameter count, and a synthetic corpus with no signal (the analysis must find nothing there).
- Everything ran on CPU.

First result on Ar-MuSA: there was almost nothing to fuse

Ar-MuSA, one seed	macro-F1
Text only	0.546
Audio only	0.418
Visual only	0.433
All three (late fusion)	0.559

Adding audio and video to text gained only **+0.013**.

Tri-CAN did not beat late fusion either, and neither did two other attention variants.

Before concluding that attention does not work, we checked whether the corpus gives it any room to work in.

Label provenance: where does the label come from?

Both corpora annotate each modality separately, plus an overall label. So we can count how often a modality's own label equals the overall label.

Ar-MuSA annotation	agrees with overall
Transcript	**96.5%**
Audio	77.1%
Frame	48.2%

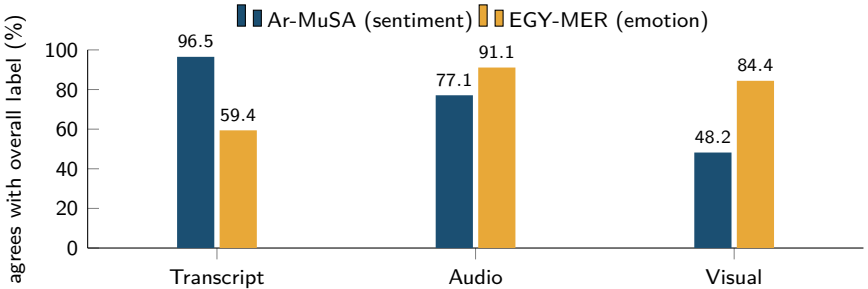
- Only 304 of 8,699 clips have an overall label different from the transcript label.
- A model that perfectly predicted the transcript label would score macro-F1 **0.966** on the "multimodal" target.
- The sarcasm flag behaves the same way (94.9%).

This is a property of the corpus. It applies to every published result on Ar-MuSA, including the attention models, not only to ours.

computed by `experiments/label_provenance.py`

The prediction, and what EGY-MER showed

If label provenance is the problem, then a corpus whose label is *not* in the text should behave differently: non-verbal modalities should matter more, and the gain from combining them should grow. We wrote this down before EGY-MER arrived.



On EGY-MER the order flips: audio carries the label best, and 40.6% of segments have an overall label different from the transcript label.

EGY-MER: 473 segments with all three modality labels

The models behaved as predicted

macro-F1, one modality	Ar-MuSA	EGY-MER
Text	**0.546**	0.340
Audio	0.418	0.477
Visual	0.433	**0.523**
All three (late fusion)	0.559	0.566
Gain over best single	+0.013	**+0.044**

- Text goes from the best modality to the worst.
- The benefit of using all three modalities is more than three times larger on EGY-MER.

So EGY-MER is a corpus where fusion *can* be tested. The next question is whether cross-modal attention does better than concatenation there.

base encoders, seed 0

Twelve fusion arrangements on EGY-MER

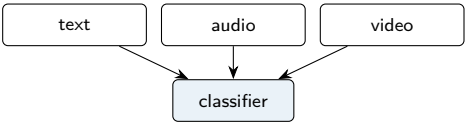
	macro-F1, mean ± SD over 3 seeds		
arrangement	base encoders	best encoders	params
late fusion (concatenation)	**0.558** ± .006	0.594 ± .009	0.9M
late fusion, widened	0.553 ± .021	0.602 ± .012	1.9M
gated fusion (published)	0.530 ± .010	0.568 ± .015	1.1M
early fusion	0.518 ± .010	0.571 ± .028	1.7M
all-pairs attention (MulT)	0.535 ± .021	0.574 ± .019	6.4M
Tri-CAN, text anchor	0.548 ± .011	0.559 ± .004	1.9M
Tri-CAN, audio anchor	0.514 ± .005	0.544 ± .010	1.9M
Tri-CAN, visual anchor	0.539 ± .017	0.554 ± .015	1.9M
hybrid, text anchor	0.541 ± .005	0.583 ± .013	2.5M
hybrid, audio anchor	0.544 ± .007	0.570 ± .024	2.5M
hybrid, visual anchor	0.545 ± .005	0.559 ± .014	2.5M
hybrid + gate, text anchor	0.557 ± .014	**0.608** ± .017	2.7M

- No arrangement beats plain concatenation by more than the seed spread, with either set of encoders.
- Concatenation is also the smallest model.
- The three Tri-CAN variants are in the bottom half both times.
- Anchoring on audio or video, EGY-MER's stronger modalities, did not help.

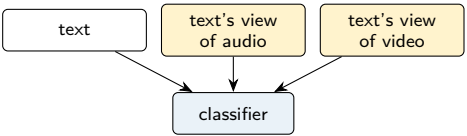
best encoders: AraBERT-twitter, wav2vec2 mid-layer, face-expression ViT

Why Tri-CAN loses: a flaw in our own design

Late fusion



Tri-CAN (text anchor)



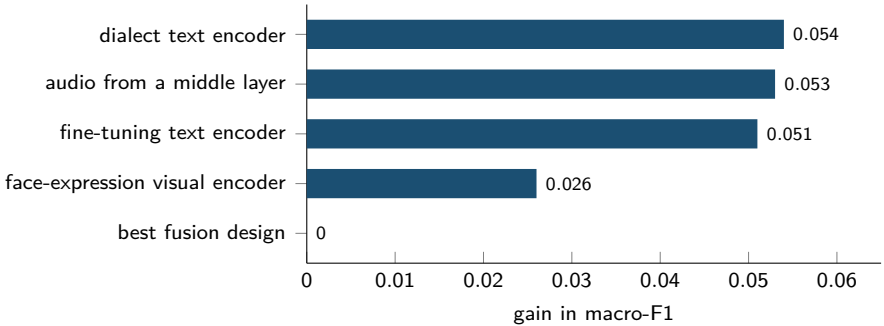
raw audio and video never reach the classifier

All three inputs to Tri-CAN's classifier are pooled over the *text* positions. So Tri-CAN does not add attention on top of late fusion; it replaces audio and video with what text finds in them, and has less information than the baseline.

On incongruent samples every anchor scores below late fusion (-0.047, -0.070, -0.066).

Repair: the "hybrid" models give the classifier the raw vectors too. With a gate added, text-anchored attention goes from -0.010 to -0.001 behind late fusion: a tie, not a gain.

Where the improvements actually came from



- Dialect text encoder: AraBERT-twitter instead of the MSA-trained AraBERTv2 (Ar-MuSA).
- Audio: the last wav2vec2 layer is tuned to its pretraining task; layer 10 carries more prosody (EGY-MER).
- Fine-tuning the text encoder raised late fusion and Tri-CAN by the same $+0.050$, so their order did not change.

Hypotheses we tested and had to reject

1. **Anchor choice.** Text is EGY-MER's weakest modality, so anchoring on audio or video should help. It did not (audio -0.034, video -0.009 against text).
2. **Multilingual audio encoder.** XLS-R scored lower than English wav2vec2-base on Ar-MuSA (0.384 vs 0.421). The weak audio was not a language mismatch.
3. **Face cropping.** Cropping to the largest detected face lowered macro-F1 by 0.038, although faces were found in 98% of frames. In talk-show footage the largest face is often not the person speaking.
4. **Emotion-trained audio encoder.** A wav2vec2 fine-tuned for English acted emotion scored lower than the generic one (0.458 vs 0.477).

From 2 and 4: matching the encoder's language and domain to the data helped; matching its training labels to our task did not.

Comparison with the published EGY-MER baseline

	weighted F1
Hussein et al. (gated late fusion)	0.519
Ours, late fusion, base encoders	0.654
Ours, late fusion, best encoders	**0.669**

ours: mean of 3 seeds, 5-fold CV

Not a like-for-like comparison

The published result uses 363 samples and its own splits. The corpus we received has 505 annotated segments; we use the 484 labelled with one of the seven emotions.

We report it because it is the only published number on this corpus, but the contribution of this project is the analysis, not this score.

Limitations

- **Small corpus.** 484 samples over 7 classes, with only 12 Fear and 18 Happiness. Macro-F1 is noisy, which is why we use seeds and call small differences ties.
- **Frozen audio and video encoders.** Only the text encoder was fine-tuned; doing the others needs a GPU we did not have.
- **Annotations are a proxy.** "The modalities disagree" means the annotators labelled them differently, not that we measured the signal directly.
- **No grouping by source video.** Folds are stratified by label but the same speaker can appear in train and test. This affects absolute scores more than the comparisons, since every model uses the same folds.
- **The corpus we received differs from the paper's** (505 annotated segments vs 363).

Conclusion

1. On Ar-MuSA, the overall label matches the transcript annotation 96.5% of the time. That leaves no room to compare fusion methods on it. This is cheap to measure and we think it should be reported with any multimodal benchmark.
2. On EGY-MER, where the label is mostly in audio and video, fusion does matter, but plain concatenation was as good as any attention design we tried.
3. The gains we did get came from choosing and using encoders well.

Next steps

- Measure label provenance on MORAD (same 7 emotions, about 6$\times$ larger).
- Active-speaker detection, so that face-based features look at the right person.
- Folds grouped by source video.

Thank you

Code, results and reports: `github.com/nourgaser/giu-ms-rp`

Backup: controls behind the null result

- **Capacity-matched baseline.** Late fusion widened to Tri-CAN's parameter count (1.9M). It does not overtake the plain version beyond seed noise, so "Tri-CAN lost because it is bigger or smaller" is ruled out.
- **Null corpus.** On synthetic data with no signal, the analysis gives chance-level scores and a difference-in-differences interval containing zero (2 automated tests).
- **Out-of-fold predictions.** Subgroup results (congruent, incongruent) are computed on predictions from models that never saw those samples.
- **Same folds, features and training loop** for every model compared.

Congruent vs incongruent, text-anchored Tri-CAN minus late fusion: $+0.036$ and -0.046; difference -0.082, 95% bootstrap interval $[-0.189, +0.034]$ (208 and 265 samples). The direction is consistent; the interval includes zero.